

ARK-463 Results — Production Deployment · Exact-Action Binding

Experiment ID: ARK-463

Execution Date: 2026-07-18

Status: EXECUTED — POST-LOCK

Verdict

 **PASS**

All success criteria met:

- ✓ C1: $g_allow_exact = 1.0000$ (≥ 0.95)
 - ✓ C2: $g_deny_mismatch_min = 1.0000$ (≥ 0.95)
 - ✓ C3: $g_margin = 1.0000$ (≥ 0.90)
 - ✓ C4: Concordance = $800/800 = 100.00\%$ ($\geq 95\%$)
 - ✓ C5: Kill-gate = 600 wrong-allows (≥ 50)
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Executive Summary

ARK-463 tested whether ExecutionProof-based deployment guards correctly enforce exact-action binding across all 5 deployment tuple dimensions (service_name, environment, version, region, deployment_method).

Key Findings:

- **Perfect exact-match enforcement:** All 200 exact-match scenarios (Arms 1, 8) correctly allowed
 - **Perfect mismatch rejection:** All 600 mismatch scenarios (Arms 2-7) correctly denied
 - **100% dual-guard concordance:** V1 (JavaScript) and V2 (Python) agreed on all 800 decisions
 - **Testbed falsifiable:** Broken guard produced 600 wrong-allows, demonstrating detection capability
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Detailed Metrics

Overall Performance

Metric	Value	Threshold	Status
Total scenarios	800	-	-
Dual-guard concordance	800/800 (100.00%)	≥95%	✓ PASS
g_allow_exact	1.0000	≥0.95	✓ PASS
g_deny_mismatch_min	1.0000	≥0.95	✓ PASS
g_margin	1.0000	≥0.90	✓ PASS

Per-Arm Performance

Arm	Description	Scenarios	V1 Correct	V2 Correct	Concordant
1	Exact Match → ALLOW	100	100/100	100/100	100/100
2	Service Name Mis-match → DENY	100	100/100	100/100	100/100
3	Environment Mismatch → DENY	100	100/100	100/100	100/100
4	Version Mis-match → DENY	100	100/100	100/100	100/100
5	Region Mismatch → DENY	100	100/100	100/100	100/100
6	Method Mismatch → DENY	100	100/100	100/100	100/100
7	Multiple Mis-match → DENY	100	100/100	100/100	100/100
8	Exact Match Stress → ALLOW	100	100/100	100/100	100/100

Kill-Gate Results

Metric	Value
DENY scenarios tested	600 (Arms 2-7)
Wrong-allows detected	600/600
Falsifiable	✅ YES (≥50 required)

Guard Implementations

V1 Guard (JavaScript)

- Exact byte-equality comparison (`===`) on all 5 dimensions
- Decision path: single conditional checking all fields
- Performance: Average ~0.1ms per decision
- Correctness: 800/800 (100.00%)

V2 Guard (Python)

- Independent implementation using string equality (`==`)
- Decision path: individual field comparisons with detailed reason building
- Performance: Average ~0.2ms per decision
- Correctness: 800/800 (100.00%)

Implementation divergence: None observed - both guards produced identical decisions on all 800 scenarios.

Example Scenarios

Exact Match (ALLOW)

```
{
  "authorized": {
    "service_name": "user-api",
    "environment": "production",
    "version": "v2.3.1",
    "region": "us-east-1",
    "deployment_method": "rolling-update"
  },
  "presented": {
    "service_name": "user-api",
    "environment": "production",
    "version": "v2.3.1",
    "region": "us-east-1",
    "deployment_method": "rolling-update"
  },
  "decision": "ALLOW"
}
```

Version Mismatch (DENY)

```
{
  "authorized": {
    "service_name": "user-api",
    "environment": "production",
    "version": "v2.3.1",
    "region": "us-east-1",
    "deployment_method": "rolling-update"
  },
  "presented": {
    "service_name": "user-api",
    "environment": "production",
    "version": "v2.3.2",
    "region": "us-east-1",
    "deployment_method": "rolling-update"
  },
  "decision": "DENY",
  "reason": "Deployment tuple mismatch: version: 'v2.3.1' vs 'v2.3.2'"
}
```

Integrity Verification

Preregistration Lock (MANIFEST.txt):

All source files were hashed before execution. Post-execution verification confirms:

-  PREREGISTRATION.md — unchanged
-  generator/scenario_generator.py — unchanged
-  verifiers/v1_guard.js — unchanged
-  verifiers/v2_guard.py — unchanged
-  run_arms.py — unchanged
-  run_killgate.py — unchanged

Result files generated:

- results/arm_[1-8]_scenarios.json — 800 test scenarios
- results/arm_[1-8]_results.json — Per-arm execution results
- results/overall_results.json — Aggregate metrics
- results/killgate_results.json — Falsifiability test results

Conclusions

1. **Exact-action binding works as designed:** Both guard implementations correctly enforce byte-exact matching across all 5 deployment dimensions, with zero tolerance for normalization, prefix matching, or fuzzy logic.
2. **Dual implementation concordance:** 100% agreement between JavaScript and Python guards demonstrates that exact string equality is deterministic and implementation-independent.
3. **Comprehensive mismatch detection:** All 6 mismatch arms (single-dimension and multi-dimension) achieved perfect denial rates, showing the guards catch substitution attempts regardless of which field(s) differ.

4. **Edge case robustness:** Arm 8 (stress test with long names, SHAs, special chars) achieved perfect ALLOW rate, confirming exact matching works with real-world deployment identifiers.
 5. **Testbed validity confirmed:** Kill-gate produced 600 wrong-allows, proving the testbed can detect broken guards that skip exact-match enforcement.
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Limitations

This experiment demonstrates **necessary but not sufficient** conditions for production deployment security:

- Does NOT test: credential validation, network security, RBAC integration, audit logging
- Does NOT validate: deployment health checks, rollback logic, service mesh integration
- Does NOT guarantee: production-readiness of the guard implementations
- Synthetic data only: no real cloud provider APIs were tested

Next steps for production use:

- Integration with real cloud APIs (AWS, GCP, Azure)
 - Audit trail and non-repudiation
 - Rate limiting and DoS protection
 - End-to-end deployment pipeline testing (ARK-463-467 series)
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Execution completed: 2026-07-18

Preregistration: See `PREREGISTRATION.md`

Source integrity: See `MANIFEST.txt`